

L1 Norm

Grid-path distance jo sparsity create karta hai

01 Opening Hook

STORY TIME

Socho tum Manhattan jaisi city mein ho jahan sirf seedhi roads hain – koi diagonal shortcut nahi. A se B jaane ke liye tumhe right chalna padega, phir upar – total blocks ginte ho. Yeh grid-distance hi hai **L1 Norm**.

02 Why Does This Concept Exist?

THE PROBLEM

Kabhi kabhi humein outliers ke against robust measurement chahiye hoti hai, ya humein model weights ko 'sparse' (kai weights exactly zero) banana hota hai feature selection ke liye. L1 norm invent hua taaki hum yeh dono kaam kar sakein – jo L2 norm nahi karta.

03 Explain Like I'm 10

SIMPLE WORDS

Socho tumhare paas coins hain – L1 norm sirf unki total ginti count karta hai (chahe woh 1 rupee ka ho ya 10 ka, bas add karo, sign ignore karke). Yeh 'seedha-saadha total' hai, kisi ko square nahi karta.

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

L1 norm sabhi components ke absolute values ko simply add kar deta hai. Yeh badi values ko utna extra penalize nahi karta jitna L2 karta hai (jo square karta hai) – isliye L1 outliers ke against zyada 'forgiving' hota hai.

05 Ab Thoda Maths

FORMULA TIME

Formula:

$\|v\|_1 = |v_1| + |v_2| + \dots + |v_n|$ – saare absolute values ka sum.

06 Visualize Everything

PICTURE IT

Vector $[3, -4]$: $L1 \text{ norm} = |3| + |-4| = 7$

07 AI Connection

WHERE AI USES THIS

Lasso Regression: L1 regularization (Lasso) model weights ko sparse banata hai – kai weights exactly zero ho jaate hain, jo automatic feature selection jaisa kaam karta hai.

Robust Loss (MAE): Mean Absolute Error loss function L1 norm use karta hai, jo outliers ke against L2 (MSE) se zyada robust hai.

08 Real Industry Example

REAL COMPANIES

ML libraries (scikit-learn): Lasso Regression directly L1 penalty use karta hai automatic feature selection ke liye.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch: ``torch.norm(v, p=1)``. L1 regularization training loop mein loss mein add hota hai: ``loss += lambda_val * sum(p.abs().sum() for p in model.parameters())``.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: L1 norm hamesha better hai L2 se.

Sahi baat: Depend karta hai use-case par - agar sparsity chahiye toh L1, agar smooth solutions chahiye toh L2 better hota hai.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: L1 norm kya hai?

A: Vector ke saare components ke absolute values ka sum.

[Intermediate] Q: L1 regularization sparsity kyun create karta hai?

A: L1 ka gradient constant hota hai (chahe weight chhota ho), isliye optimization weights ko push karke exactly zero tak le ja sakta hai.

[Advanced] Q: MAE loss (L1-based) kab prefer karte hain MSE (L2-based) se?

A: Jab dataset mein outliers zyada hain aur unhe utna heavily penalize nahi karna chahte - MAE zyada robust hota hai un cases mein.

12 Exercises

PRACTICE TIME

[Easy] $[1, -2, 3]$ ka L1 norm nikalo.

Solution: $1+2+3 = 6$.

[Medium] $[0, 0, 5]$ ka L1 norm kya hoga?

Solution: 5.

[Hard] Lasso regression mein agar regularization strength (λ) bahut bada kar do, kya hoga?

Solution: Zyada weights exactly zero ho jaayenge – model bahut sparse (simple) ho jaayega, underfitting ka risk badh sakta hai.

13 Golden Summary

Concept	Easy Meaning
L1 Norm	Absolute values ka sum
Lasso	L1-regularized regression, sparse weights
MAE	L1-based loss function, outlier-robust
Sparsity	Kai weights exactly zero ho jaana
Use case	Feature selection, robust regression

14 One-Line Memory Trick

“L1 norm sirf seedhe-saadhe blocks ginta hai – grid path distance.”